Challenge Brief

Template for challenge partners · Since AI Hackathon 2026 · 6–8 Nov 2026, Turku

Official Europress Group Oy challenge brief for Since AI Hackathon 2026. The challenge has two connected parts: obtaining affordable compactor-weight data and turning it into measurable customer and business value.

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Challenge partner (company)

Europress Group Oy

01 Challenge Title

A short, memorable name for this two-part industrial challenge.

SMART Material Value Engine — Weigh at Source, Create Value

02 Problem Statement

Explain the current problem and why both measurement and commercial value matter.

Today, waste compactor fill levels are known, but the commercially decisive material weight often becomes available only when the waste operator weighs a load at a processing facility. That weight determines charges and credits, leaving the operator in a strong position to control the data. Europress wants usable, independent weight information already where waste is generated.

PART 1 — GET THE WEIGHT:
How could Europress obtain sufficiently accurate weight data directly at the compactor with an affordable and scalable solution? Solution could be integrated into or retrofitted to a compactor. Explain sensor choice, installation, calibration, limitations, costs and a validation experiment.
PART 2 — CREATE VALUE FROM THE WEIGHT: Once the weight data is available at the source, explore how Europress could combine it with SMART and commercial data to create measurable customer value and new commercial services. Demonstrate at least one concrete use case, such as €/t optimisation, material-value transparency, anomaly detection or recommended actions, but also consider other services, business models and revenue opportunities enabled by independent weight data.

03 Guiding Question

One question connecting the two challenge parts.

How can Europress measure waste weight affordably at the source and turn independent weight data into measurable customer value, new SMART services and commercial opportunities?

04 Recommended AI Approaches *Optional*

Possible sensing principles, analytical tools or AI methods. No single technique is required.

PART 1: Compare feasible sensing options (e.g., load cells, structure/strain measurements, hydraulic signals or a justified alternative). Explain placement, accuracy target, expected cost, retrofitting, calibration and an experiment to validate the approach; no live-machine integration is required.

PART 2: Explore AI and analytics approaches such as anomaly detection, predictive models, pattern recognition and recommendation models to identify new value from weight and SMART data. Compare AI-driven insights with transparent rule-based logic where relevant, and consider explainability, uncertainty and human approval of recommended actions.

05 Dataset or API Provided *If any*

Describe the supplied simulated dataset and what it cannot validate.

> At Since AI, challenges can be built with synthetic, anonymized or public data — no personal data is required.

Since AI provides a synthetic, final quickstart and competition dataset (CSV/Parquet): approximately 500 fictional compactors across a 365-day simulation, including weight observations, fill levels, collections, weighbridge records, illustrative contracts, costs, invoices and anomalies. The supplied data supports PART 2; it does NOT prove or benchmark weighing hardware for PART 1. No proprietary Europress data, test rigs or live APIs are provided.

06 Evaluation Criteria

Use a balanced assessment of sensing feasibility and business value.

25% weighing concept (technical feasibility, accuracy reasoning, cost and retrofit potential);
25% customer/business value;
20% data-driven value concept and analytical reasoning (including traceable calculations and/or other measurable value enabled by weight data)
15% demonstrated software workflow and SMART integration pathway;
10% commercial productisation potential;
5% clarity, uncertainty and practical limitations.

07 Submission Format

State the practical concept and software deliverables expected during the hackathon.

Submit via Since AI: (1) a short illustrated weighing concept describing sensor/measurement principle, placement, indicative bill of materials and cost, installation, calibration, error sources and validation plan; (2) a working software demo or recorded

walkthrough using synthetic weight data, with code/setup notes, traceable economics and a proposed SMART integration; (3) a concise customer value and business case describing the proposed use case or service, measurable customer benefit, potential commercial model for Europress and how the solution could be productised as part of SMART. Identify every third-party library, model, dataset or service used, its licence and any relevant commercial-use restrictions. Include assumptions and what remains untested. No physical prototype required.

08 Point of Contact

Contact details for technical and event-related questions.

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